

MorningStar-Lab v1.0 — Riemann Hypothesis Reconnaissance Certificate

100 nontrivial zeros, all `RH_ok=True`, $\beta_{\text{KMS}} = 2.0$

Entangled Technologies — The Morning Star Project
Genesis: `eecbcd9a...875f` Axiom Debt: `[]`

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Scope (honest)

This certificate records a *numerical reconnaissance* of the Riemann ζ -function on the critical line $\Re(s) = \frac{1}{2}$, performed by the MorningStar-Lab `kernel.py` probe via an arbitrary-precision `mpmath` backend at `workdps=50`. It is **not** a proof of the Riemann Hypothesis. The Lean 4 axiom-debt-`[]` result that ships in `lean-proof/` concerns the M1–M10/M13 BC–CM ($h = 1$) spine and is *independent* of this document.

Protocol

For $s = 0.5 + it$ the kernel defines `RH_ok` $\equiv [|L(s)| < 10^{-12}]$ and $\beta_{\text{KMS}} = 1/\Re(s) = 2$ on the critical line. Each probe is appended to the Genesis-sealed ledger `data/hits.txt` with a nanosecond timestamp and a SHA-256 covering the canonical line body. Because the timestamp is part of the hash input, two probes at the same point still receive distinct ledger SHAs: every measurement is uniquely receipted.

Genesis seal

```
Genesis SHA-256 (preamble, lines 1--9 of data/hits.txt):
  eecbcd9a540aa7a2c90edd23827c73e4d1bb5af641d352f70a5de849b21f875f
Genesis lines 5--9 (frozen):
437
1094
axioms=[] 2026-05-24
M13_CERT_SHA256=d99b0df4b6e3a5817a90e7f8da44b20261d3b670d31309779a776efbf1bcc668
--- GENESIS SEAL ---
```

Lean axiom debt

```
sha256(lean-proof/VERIFY.txt) =
4150d231bb01d1bca0ecd73093002b167be1f4e52439e1dc307fb8f9fdf1bfd8
```

The frozen M1–M10/M13 spine compiles in `lean-proof/` with zero axioms. The Genesis-pinned `M13_CERT_SHA256=d99b0df4...` is the sealed M13 certificate value at v1.8-BC release time and is *not* expected to equal the current `VERIFY.txt` hash — the latter is regenerated by `lean-proof/regenerate.sh`; the Genesis line is immutable.

Results: first 100 nontrivial zeros (MPMATH_ZETA)

Produced by `python -c "import kernel; kernel.hunt_zeros(1, 100)"`. Of the 100 probes, **100** returned `RH_ok=True`; `kms.beta` is uniformly 2.0: **true**; `tag` is uniformly `MPMATH_ZETA: true`. $|L|$ values shown are the float-rounded probe value (the kernel passes `float(zetazero(n)).imag` into `probe()`, so $|L|$ is $\sim 10^{-15}$ rather than 0; this is honestly recorded).

n	t_n (mpmath, 20 sig.fig.)	$ L(0.5 + it_n) $	ledger SHA (16 hex)
1	14.13472514173469379	6.6681863422837782815e-16	bf9120b732c1472a...
2	21.022039638771554993	1.1609164044328830424e-15	1854b72d2d97a66d...
3	25.010857580145688763	8.4988448557528602897e-16	cf6615f05ac9a15c...
4	30.42487612585951321	1.0578057324100325666e-15	8b2c11430341469b...
5	32.935061587739189691	2.7490910092466363602e-15	4d6c5bcc279ed803...
6	37.586178158825671257	6.6669427104052746089e-15	329e0f38923f908b...
7	40.918719012147495187	4.7945815205315569849e-15	70f2e0c30e51cd40...
8	43.327073280914999519	4.316709177172472642e-15	8b3e83fc8d304509...
9	48.005150881167159728	1.99980780138915588e-15	8e53c1dab0dba7dd...
10	49.773832477672302182	2.4488081356777964299e-15	b61dd4380a245d74...
11	52.970321477714460644	7.5456425078431052624e-15	fbaa7ef82372a0f0...
12	56.446247697063394804	7.7937830463317696222e-15	59a79c3ca3d3bc5f...
13	59.34704400260235308	8.5717340578274243528e-16	a7d08708b9eaa0aa...
14	60.831778524609809844	2.0252763203965030083e-16	f86c1db1fb8e72f8...
15	65.112544048081606661	9.6228941372206140762e-15	7995ecbf75fd6b27...
16	67.079810529494173714	1.0582484029808909341e-14	a153284e45d73cb0...
17	69.546401711173979253	1.2454939415804993309e-14	58122affe8f911aa...
18	72.067157674481907583	7.9364022008164876089e-15	deb1c1a76423abb4...
19	75.704690699083933168	1.2543277824394555479e-14	9cb879159846d89d...
20	77.144840068874805373	8.5609983118655008102e-15	85d0484627ed55ec...
21	79.337375020249367923	6.3291105470374835166e-16	a4c23451cb8b0dec...
22	82.910380854086030183	4.3301347212692145827e-15	f3ee2796b535620e...
23	84.735492980517050106	2.7213572575263907203e-15	4a5f245df0864b55...
24	87.425274613125229407	7.5968399156873348862e-15	6d35f982585d3bbe...
25	88.809111207634465424	1.473998180376794053e-14	c1b051e01731e780...
26	92.491899270558484296	2.0905025173378787494e-14	e77e360d224fee34...
27	94.651344040519886967	1.2513987259314612489e-15	48d7fe1f98234141...
28	95.870634228245309759	3.1561500045397633862e-15	64c1a5f2fd8ff97a...
29	98.831194218193692233	1.8024448817653808952e-14	0b627f34e2028e1a...
30	101.31785100573139123	2.1663068099031809468e-14	3f0db0ef04c48ad9...
31	103.72553804047833942	3.2537612149115055821e-15	22a54c84d98f370c...
32	105.44662305232609449	1.0275589438875036232e-14	63b710292e5f6702...
33	107.16861118427640752	2.0476252063089412951e-14	e93737fe4d7c5d04...

n	t_n (mpmath, 20 sig.fig.)	$ L(0.5 + it_n) $	ledger SHA (16 hex)
34	111.02953554316967452	4.0225258199627817264e-15	a19b0ba63fb71aa5...
35	111.87465917699263709	3.2851501507094203876e-16	5d80b1b88ea1abd4...
36	114.32022091545271277	1.1486916741746918214e-14	f3256d436c84209e...
37	116.22668032085755438	1.6536827984576063967e-15	8da72b788e361790...
38	118.79078286597621732	2.1766507058517601639e-14	8137e10bb0fe4a71...
39	121.37012500242064592	9.737768664731728901e-15	7fce29c00703f8ce...
40	122.9468292935525882	9.0676774017799263968e-15	f1b855f45c37a235...
41	124.25681855434576718	6.9066195628086063049e-15	02e9cdf73a5a1d15...
42	127.51668387959649512	1.60359041461546306e-14	86542305ca085d2c...
43	129.57870419995605099	3.1109366910934847433e-14	d3ee81d5527e0d5c...
44	131.08768853093265672	2.3161276364618314847e-14	0ff7325451bfd592...
45	133.49773720299758645	2.6808010613461243083e-14	b0202f693a84b741...
46	134.75650975337387133	1.4541303509256516328e-14	83ad76bee2c24ad5...
47	138.1160420545334432	1.7941977049347236653e-14	dc601c7747ca3d71...
48	139.73620895212138895	2.643243585531812547e-15	c8e465b118741a47...
49	141.12370740402112376	1.7757712450187357658e-14	9b439d93797d5184...
50	143.11184580762063274	2.8131536393153893442e-14	829dce85a8c47fe5...
51	146.00098248676551855	3.3011566497020035363e-14	523a4325fdffea0f...
52	147.42276534255960205	3.5103075082861796569e-14	258348eaea1b3874...
53	150.05352042078488035	4.065077765218085594e-15	2e679646c0a5885e...
54	150.92525761224146676	2.0619727226471015361e-16	d70581de0c303b80...
55	153.0246938111988962	3.9842227512548867114e-14	802567041c3b230c...
56	156.11290929423786757	3.9814929106252461136e-14	faa9882320ef9307...
57	157.59759181759405989	8.6990252604351729143e-15	4bb47ea76b917a8a...
58	158.84998817142049872	1.5492464966808554115e-14	ae6f78937a7f270e...
59	161.18896413759602752	1.222227754479819105e-14	cc47b4176754140e...
60	163.03070968718198724	3.8145522877079241274e-14	7d909e3f44a3a75e...
61	165.53706918790041883	1.5934421258016105164e-14	7740638d48ee681f...
62	167.18443997817451344	9.351647075368275422e-15	9fd98243f3ba143c...
63	169.09451541556882149	1.2406623346166143728e-17	c02a8e106c44ef0b...
64	169.91197647941169897	1.4185805788867766762e-14	99bd8409f3cc57bf...
65	173.41153651959155296	9.8600385153570357148e-15	57df70bb3abf02a9...
66	174.75419152336572581	1.8039923679934819799e-14	85650672abb91366...
67	176.44143429771041889	2.8868823733470803716e-14	2a04b5ac031f7648...
68	178.37740777609997729	1.583935256339428881e-14	cc81908dcc6587de...
69	179.91648402025699614	2.0303515076822262535e-14	0dbc01e1b79b2219...
70	182.20707848436646192	3.3781472479752955672e-15	5d8c760e98c0a949...
71	184.8744678483875088	1.859042863665181302e-14	742b28babdf58219...
72	185.59878367770747147	2.281135853745856043e-15	3aa18e82f6f82875...
73	187.22892258350185199	1.0147017343537332008e-14	c1cb8982f8b51a8e...
74	189.41615865601693708	2.2724719150009087227e-14	7a728657871d08c3...
75	192.02665636071378655	8.4676046329067046506e-16	dce3ca119ffc7321...
76	193.07972660384570405	1.0564189038469640047e-14	19b3e08c66901d81...
77	195.26539667952923532	9.5798719635808800325e-15	2b01cba6f16ee32e...
78	196.87648184095831695	6.5144464987030002029e-15	520aeb76747d0019...
79	198.01530967625191242	1.4214137402893191203e-14	19fcff55cd60e70f...

n	t_n (mpmath, 20 sig.fig.)	$ L(0.5 + it_n) $	ledger SHA (16 hex)
80	201.26475194370378873	3.7950730241263233035e-14	151b9bee1f604d4a...
81	202.49359451414053428	1.2301130737889190311e-14	53b8a0e8bcc8cd41...
82	204.18967180310455433	1.8888765559417519343e-14	bc8588bea9843f30...
83	205.39469720216328603	1.6841363552102820469e-16	1e70515ad78ce05e...
84	207.90625888780620986	3.322584032230280077e-14	c4762b855090ddf2...
85	209.57650971685625985	2.008894539759053534e-14	c5a35ae7c3b97b8f...
86	211.69086259536530756	1.5364423528023801215e-14	f310f7ff00accc24...
87	213.34791935971266619	2.1754896273293151388e-14	b8faf10f9ea05053...
88	214.54704478349142322	1.4085147395161649892e-14	a701fb11d0c56597...
89	216.16953850826370027	5.6583924185142960164e-14	e3faac81bea4cbf1...
90	219.06759634902137899	2.8972620826254159942e-14	f8fe964ba05b01c4...
91	220.71491883931400337	8.822851837811300474e-15	f82d21f7f70f5f33...
92	221.4307055469333873	9.7422628106292878048e-15	67d3c1f2fad1de33...
93	224.00700025460433521	3.683769192119255174e-14	887bced19b542860...
94	224.9833246695822875	1.2150103989842640456e-15	f4a77e8eaf25d870...
95	227.42144427967929131	5.2034436411560927069e-15	d45b405c52899eb2...
96	229.33741330552534811	4.2035945231418680698e-14	7a5503aa28e52fb1...
97	231.25018870049916477	1.6085249554601252422e-15	da8376317370dd45...
98	231.9872352531802486	5.5009747488587620294e-15	9d879e1d4e1ea03b...
99	233.69340417890830064	3.8826989082892929941e-14	d51ea3dfc40a59b3...
100	236.5242296658162058	4.9762038068635896212e-14	52047b3b24ae8cac...

Bracket verification: `bracket_zero(1, 1e-06)`

Tight sweep of $[t_1 - 10^{-6}, t_1 + 10^{-6}]$ at step $2 \cdot 10^{-7}$ through `scan_critical_line(1, ..., h=1)`.
Float arithmetic visits 1 point(s) within the gunsight ($|L| < 10^{-12}$):

$\Im(s)$ probed	$ L $	ledger SHA (16 hex)
14.134725141734695	6.6681863422837782815e-16	1dc2a711bcc4fe51...

$t_1 \approx 14.134725141734695$, window = $1e - 06$, step = $2e - 07$.

Reproducibility

```
# Verify the seal:
python scripts/check-genesis-seal.py

# Regenerate this certificate (re-runs hunt_zeros(1, 100) and bracket_zero):
python scripts/generate-rh-cert.py

# Full 7-step validation harness (requires 'lake' on PATH for Lean):
bash scripts/validate-morningstar.sh
```

What this certificate does not claim

- It does **not** claim a proof of RH. `RH_ok=True` on a probe means $|L(s)| < 10^{-12}$ at one finite-precision sample, nothing more.
- The `MPMATH_ZETA` backend is not Lean-verified. For $h \geq 2$ the kernel logs `NEEDS_SAGE` with `reason=h>=2_out_of_scope_for_mpmath_backend`; those rows carry a deterministic stub `L_nonvanish=True` and must not be read as L-function evaluations.
- The names `hit_437` and `hit_1094` in `lean-proof/TheoremaAureum/AutoLemmas.lean` are tautologies (`True := trivial`) referencing the OpenCV cube counts from README Appendix A; they claim nothing about number theory.